THE CONTINUOUS K-GROUPS OF NUCLEAR $\mathbb{C}_P$-VECTOR SPACES

WEI YANG

ABSTRACT. We compute the homotopy groups of $K\big(\mathrm{Nuc}(\mathbb{C}_p)\big)$, together with their Galois-equivariant and condensed refinements. They vanish in negative and positive even degrees, while $K_0 \cong \mathbb{Z}$, $K_1 \cong \mathbb{C}_p^\times$, and, for $k \geq 2$,

$$K^{\mathrm{cont}}_{2k-1}\big(\mathrm{Nuc}(\mathbb{C}_p)\big) \cong \bigoplus_{\ell \neq p} (\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k) \oplus (B^+_{\mathrm{cris}})^{\varphi=p^k}/\mathbb{Z}_p(k).$$

This high-odd isomorphism may be chosen for the underlying abstract Galois action. Its prime-to-p splitting is induced by reduction to the special fiber, whereas the crystalline identification is generally noncanonical. In condensed abelian groups we retain the internal continuous Galois action and identify the p-primary term with Fontaine's quotient topology. The proof uses formal/Tate localization, finite-coefficient rigidity, the rational Beilinson comparison, and the arithmetic gluing map supplied by the cyclotomic trace.

1. INTRODUCTION

Let $\mathbb{C}_p$ be the completion of $\overline{\mathbb{Q}}_p$, let $\mathcal{O}_{\mathbb{C}_p}$ be its valuation ring, and write $G_{\mathbb{Q}_p} = \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$. Here $\mathrm{Nuc}(\mathbb{C}_p)$ is the stable category of nuclear $\mathbb{C}_p$-vector spaces in the sense of Clausen–Scholze. Dahlhausen–Yaylali–Zhou identify $K\big(\mathrm{Nuc}(\mathbb{C}_p)\big)$ canonically with the spectrum-valued continuous K-theory of the Tate ring $\mathbb{C}_p$ [6, Corollary 4.8]. Thus

$$K^{\mathrm{cont}}_n\big(\mathrm{Nuc}(\mathbb{C}_p)\big) := \pi_n K\big(\mathrm{Nuc}(\mathbb{C}_p)\big)$$

is the homotopy group of a spectrum, not of a pro-spectrum. We compute these groups and then determine the structure retained by the Galois action on $\mathbb{C}_p$.[1]

The symbols B^+_{cris} and B^+_{dR} denote Fontaine's positive crystalline and de Rham period rings, and φ denotes crystalline Frobenius. For a prime ℓ and $k \geq 1$, put

$$(\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k) := \operatorname*{colim}_a \mu_{\ell^a}(\mathbb{C}_p)^{\otimes k}.$$

We also use

$$\mathbb{Z}_p(1) = \varprojlim_r \mu_{p^r}(\mathbb{C}_p), \qquad \mathbb{Z}_p(k) = \mathbb{Z}_p(1)^{\otimes k}, \qquad \mathbb{Q}_p(k) = \mathbb{Z}_p(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p,$$

where the transition maps are the pth-power maps. Fontaine's fundamental sequence fixes the inclusion used below:

$$0 \longrightarrow \mathbb{Q}_p(k) \longrightarrow (B^+_{\mathrm{cris}})^{\varphi=p^k} \longrightarrow B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}} \longrightarrow 0.$$

We regard $\mathbb{Z}_p(k) \subset \mathbb{Q}_p(k)$ as a subgroup of $(B^+_{\mathrm{cris}})^{\varphi=p^k}$. Direct sums indexed by ℓ are over prime numbers.

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[1] The group calculation was stated in Efimov's recorded lecture [8]. The bibliography cites its MathNet page because the previously circulated YouTube link is no longer accessible.

Theorem 1.1. *There are isomorphisms of ordinary abelian groups*

$$K_0^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{Z}, \qquad K_1^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{C}_p^{\times},$$

and

$$K_n^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = 0 \qquad \text{if } n < 0 \text{ or if } n > 0 \text{ is even.}$$

For every $k \geq 2$ there is an isomorphism

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \left(\bigoplus_{\ell \neq p} (\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\right) \oplus \frac{(B_{\mathrm{cris}}^{+})^{\varphi = p^k}}{\mathbb{Z}_p(k)}.$$

The last direct-sum decomposition exists in **Ab** *but is in general noncanonical.*

The action of $G_{\mathbb{Q}_p}$ on $\mathbb{C}_p$ induces an action on $K(\mathrm{Nuc}(\mathbb{C}_p))$ by scalar twisting. Write $G_{\mathbb{Q}_p}^{\delta}$ when its profinite topology is forgotten.

Theorem 1.2 (Galois-equivariant homotopy groups)**.** *The identifications*

$$K_0^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{Z}, \qquad K_1^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{C}_p^{\times}$$

are $G_{\mathbb{Q}_p}$-equivariant when $\mathbb{Z}$ has the trivial action and $\mathbb{C}_p^{\times}$ its natural action. All negative and positive even groups are zero as Galois modules.

For every $k \geq 2$, localization gives a split short exact sequence in $\mathbf{Ab}^{G_{\mathbb{Q}_p}^{\delta}}$

$$0 \longrightarrow \bigoplus_{\ell \neq p} (\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k) \longrightarrow K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \longrightarrow \pi_{2k-1}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))_{(p)}\big) \longrightarrow 0. \tag{1}$$

Consequently, after choosing the isomorphism of Theorem 5.3, there is an isomorphism of abstract $G_{\mathbb{Q}_p}$-modules

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \left(\bigoplus_{\ell \neq p} (\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\right) \oplus \frac{(B_{\mathrm{cris}}^{+})^{\varphi = p^k}}{\mathbb{Z}_p(k)}. \tag{2}$$

The splitting of (1) *is induced by reduction to $\mathcal{O}_{\mathbb{C}_p}/p$. The final crystalline identification in* (2) *is generally noncanonical.*

The preceding theorem remembers the Galois action but not its topology. Fix an uncountable strong-limit cardinal κ large enough for $\mathbb{C}_p$ and $G_{\mathbb{Q}_p}$. For an abstract abelian group A, let ΔA denote its discrete realization in $\mathrm{Cond}_\kappa(\mathbf{Ab})$. For a topological abelian group T, write

$$\underline{T}^{\mathrm{top}}(S) = \mathrm{Cont}(S, T)$$

on profinite test objects. The continuous action on $\mathbb{C}_p$ gives the condensed spectrum $\underline{K}^{\mathrm{nuc}}(\mathbb{C}_p)$ an internal action of the condensed profinite group $\underline{G}_{\mathbb{Q}_p}$.

For $k \geq 2$, give $(B_{\mathrm{cris}}^{+})^{\varphi = p^k}$ Fontaine's closed-subspace Banach topology from B_{cris}^{+}. Fontaine proves that it is also closed in B_{dR}^{+}, with the same induced topology, and that

$$0 \longrightarrow \mathbb{Q}_p(k) \longrightarrow (B_{\mathrm{cris}}^{+})^{\varphi = p^k} \longrightarrow B_{\mathrm{dR}}^{+}/\operatorname{Fil}^k B_{\mathrm{dR}}^{+} \longrightarrow 0$$

is strict, continuous, and $G_{\mathbb{Q}_p}$-equivariant [9, Section 3.1].

Theorem 1.3 (Condensed Galois-equivariant homotopy groups)**.** *For every $k \geq 2$, there is an isomorphism in*

$$\mathrm{Mod}_{\mathbb{Z}[\underline{G_{\mathbb{Q}_p}}]}(\mathrm{Cond}_\kappa(\mathbf{Ab}))$$

of the form

$$\pi_{2k-1}^{\mathrm{cond}}\underline{K}^{\mathrm{nuc}}(\mathbb{C}_p) \cong \bigoplus_{\ell \neq p} \Delta\big((\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\big) \oplus \operatorname{coker}\left(\underline{\mathbb{Z}_p(k)}^{\mathrm{top}} \longrightarrow \underline{(B_{\mathrm{cris}}^+)^{\varphi=p^k\,\mathrm{top}}}\right). \tag{3}$$

The prime-to-p splitting is canonical relative to reduction to the special fiber. With the comparison coordinates used in the proof, the identification of the p-local summand with the displayed cokernel is noncanonical: it permits a scalar in $\mathbb{Q}_p^\times$ on the rational period coordinate. Equation (3) *is not asserted to arise from a splitting of condensed spectra.*

The cokernel in Theorem 1.3 has the expected quotient topology: canonically and internally $G_{\mathbb{Q}_p}$-equivariantly, it is

$$\underline{(B_{\mathrm{cris}}^+)^{\varphi=p^k}/\mathbb{Z}_p(k)}^{\mathrm{top}}.$$

The noncanonicity in the theorem concerns the K-theoretic comparison, not this topological quotient.

Two independent calculations enter the integral proof. For every $m \geq 2$, finite-coefficient rigidity gives

$$\pi_{2i}(K(\mathrm{Nuc}(\mathbb{C}_p))/m) = \mathbb{Z}/m(i), \qquad \pi_{2i+1}(K(\mathrm{Nuc}(\mathbb{C}_p))/m) = 0 \quad (i \geq 0).$$

This determines torsion and divisibility but does not exclude a rational vector space in an integral even degree. The rational Beilinson fiber square supplies the missing information:

$$\pi_{2i}((K(\mathrm{Nuc}(\mathbb{C}_p)))_{\mathbb{Q}}) = 0, \qquad \pi_{2i-1}((K(\mathrm{Nuc}(\mathbb{C}_p)))_{\mathbb{Q}}) = B_{\mathrm{dR}}^+/\operatorname{Fil}^i B_{\mathrm{dR}}^+ \quad (i \geq 2).$$

The odd finite-coefficient vanishing makes every positive even integral group torsion-free, and the rational vanishing then forces it to be zero. In high odd degrees, the Bockstein sequence reconstructs the torsion and the rational period quotient. Fontaine's sequence rewrites the p-primary contribution in crystalline form.

The equivariant argument requires the intervening extension class, which is not determined by its finite and rational end terms. The cyclotomic-trace square of [1, (23)] identifies the p-local gluing morphism, while reduction to the special fiber supplies the prime-to-p retraction. For the condensed theorem, these comparisons are made naturally on all profinite test objects; strictness of Fontaine's sequence then identifies the cokernel with the topological quotient.

Section 2 establishes the formal/Tate model and its low-degree consequences. Section 3 computes the finite, rational, and support terms, and Section 4 performs the integral Bockstein reconstruction. Sections 5 and 6 prove the abstract and condensed equivariant refinements.

2. The formal/Tate model

Convention 2.1. The symbol $K(A)$ denotes nonconnective algebraic K-theory. For a spectrum E, write $E/m = E \wedge \mathbb{S}/m$ and $E_{\mathbb{Q}} = E \wedge H\mathbb{Q}$. All inverse limits of spectra are homotopy inverse limits. The notation $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$ means the formal K-theory spectrum

$$K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) := \operatorname*{holim}_{r \geq 1} K(\mathcal{O}_{\mathbb{C}_p}/p^r).$$

The notation $K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p)$ means the K-theory spectrum with support at p:

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) := K(\mathcal{O}_{\mathbb{C}_p} \text{ on } (p)) \simeq \operatorname{fib}\big(K(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathbb{C}_p)\big).$$

The filtration on B_{dR}^{+} is the usual decreasing filtration; for $\mathbb{C}_p$, $\mathrm{Fil}^k B_{\mathrm{dR}}^{+} = t^k B_{\mathrm{dR}}^{+}$ after choosing compatible p-power roots of unity.

Proposition 2.2 (Formal/Tate pushout)**.** *There is a bicartesian square of spectra*

$$\begin{array}{ccc} K(\mathcal{O}_{\mathbb{C}_p}) & \longrightarrow & K(\mathbb{C}_p) \\ \downarrow & & \downarrow \\ K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) & \longrightarrow & K\big(\mathrm{Nuc}(\mathbb{C}_p)\big). \end{array}$$

Equivalently,

$$K\big(\mathrm{Nuc}(\mathbb{C}_p)\big) \simeq K(\mathbb{C}_p) \coprod_{K(\mathcal{O}_{\mathbb{C}_p})} K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}), \tag{4}$$

$$K\big(\mathrm{Nuc}(\mathbb{C}_p)\big) \simeq \mathrm{cofib}(K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \longrightarrow K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})). \tag{5}$$

Proof. The ring $\mathcal{O}_{\mathbb{C}_p}$ is a characteristic-zero valuation domain. Thus p and all its powers are non-zero-divisors, $\mathbb{C}_p = \mathcal{O}_{\mathbb{C}_p}[1/p]$, and $\mathcal{O}_{\mathbb{C}_p}$ is p-adically complete. Since $\mathcal{O}_{\mathbb{C}_p}$ is p-torsion-free, its classical p-completion agrees with its derived p-completion. The principal ideal (p) is weakly proregular: for a principal ideal (a) this is equivalent to bounded a^∞-torsion, and the p^∞-torsion of $\mathcal{O}_{\mathbb{C}_p}$ is zero.

Dahlhausen–Yaylali–Zhou's extension of Efimov continuity [6, Theorem 4.6] therefore gives

$$K\big(\mathrm{Nuc}(\mathcal{O}_{\mathbb{C}_p})\big) \simeq \operatorname*{holim}_r K(\mathcal{O}_{\mathbb{C}_p}/p^r) = K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}).$$

Efimov's general formula [7, Corollary 7.10] first uses Koszul derived quotients, while weak proregularity identifies their pro-system with the ordinary quotients. In the present case the Koszul quotients are discrete because

$$H^{-1}\,\mathrm{Kos}(\mathcal{O}_{\mathbb{C}_p}; p^r) = \mathrm{Ann}_{\mathcal{O}_{\mathbb{C}_p}}(p^r) = 0, \qquad H^0\,\mathrm{Kos}(\mathcal{O}_{\mathbb{C}_p}; p^r) = \mathcal{O}_{\mathbb{C}_p}/p^r.$$

The nuclear Tate-localization diagram of [6, Lemma 4.7 and Corollary 4.8] is a localization diagram of dualizable stable categories. Applying the localizing invariant K gives the displayed square. Algebraic localization gives

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \longrightarrow K(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathbb{C}_p).$$

Pushing this sequence out along $K(\mathcal{O}_{\mathbb{C}_p}) \to K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$ gives (5). $\square$

Remark 2.3. Kerz–Saito–Tamme's original continuous K-theory is pro-spectrum-valued. Proposition 2.2 concerns the spectrum-valued theory of nuclear categories, equivalently the actual inverse-limit realization in this example.

2.1. **Low and negative degrees.** We compute the low homotopy groups of the formal term, including the derived-limit terms in degrees zero and one.

Lemma 2.4. *For every $r \geq 1$,*

$$K_0(\mathcal{O}_{\mathbb{C}_p}/p^r) \cong \mathbb{Z}, \qquad K_n(\mathcal{O}_{\mathbb{C}_p}/p^r) = 0 \quad (n < 0).$$

Proof. The ring $\mathcal{O}_{\mathbb{C}_p}/p^r$ is local, so every finitely generated projective $\mathcal{O}_{\mathbb{C}_p}/p^r$-module is free and $K_0(\mathcal{O}_{\mathbb{C}_p}/p^r) = \mathbb{Z}$.

Reduction modulo p^r is unchanged by p-adic completion, and every algebraic element lies in a finite extension of $\mathbb{Q}_p$. Therefore

$$\mathcal{O}_{\mathbb{C}_p}/p^r \cong \mathcal{O}_{\overline{\mathbb{Q}}_p}/p^r \cong \operatorname*{colim}_{L/\mathbb{Q}_p \text{ finite}} \mathcal{O}_L/p^r. \tag{6}$$

Each $\mathcal{O}_L/p^r$ is an Artinian local ring, so its negative K-groups vanish [15, Chapter III, Exercise 4.3]. Nonconnective K-theory preserves filtered colimits [2, Definition 9.6, Lemma 9.7, and

Theorem 9.8]; perfect complexes and their morphisms over a filtered colimit of rings descend to a finite stage. Applying this to (6) proves $K_n(\mathcal{O}_{\mathbb{C}_p}/p^r) = 0$ for $n < 0$. $\square$

Lemma 2.5. *The transition maps*

$$K_i(\mathcal{O}_{\mathbb{C}_p}/p^{r+1}) \longrightarrow K_i(\mathcal{O}_{\mathbb{C}_p}/p^r)$$

are surjective for $i = 1, 2$. Moreover,

$$\lim_r K_1(\mathcal{O}_{\mathbb{C}_p}/p^r) \cong \mathcal{O}_{\mathbb{C}_p}^\times.$$

Proof. Because $\mathcal{O}_{\mathbb{C}_p}/p^r$ is commutative and local, $K_1(\mathcal{O}_{\mathbb{C}_p}/p^r) = (\mathcal{O}_{\mathbb{C}_p}/p^r)^\times$. The kernel of $\mathcal{O}_{\mathbb{C}_p}/p^{r+1} \to \mathcal{O}_{\mathbb{C}_p}/p^r$ is square-zero, so every unit lifts. This proves surjectivity for K_1. Completeness of $\mathcal{O}_{\mathbb{C}_p}$ gives $\mathcal{O}_{\mathbb{C}_p} \cong \lim_r \mathcal{O}_{\mathbb{C}_p}/p^r$, and a compatible element is a unit precisely when its image modulo p is a unit. Hence $\mathcal{O}_{\mathbb{C}_p}^\times \cong \lim_r (\mathcal{O}_{\mathbb{C}_p}/p^r)^\times$.

For K_2, the residue field of every $\mathcal{O}_{\mathbb{C}_p}/p^r$ is the infinite field $\overline{\mathbb{F}}_p$. The Dennis–Stein–van der Kallen presentation for a commutative local ring with residue field of cardinality greater than five identifies K_2 with the corresponding Milnor-symbol presentation; see [11, Proposition 3.6(ii),(iii)]. The surjection $\mathcal{O}_{\mathbb{C}_p}/p^{r+1} \to \mathcal{O}_{\mathbb{C}_p}/p^r$ lifts units, so it lifts every symbol generator. Therefore $K_2(\mathcal{O}_{\mathbb{C}_p}/p^{r+1}) \to K_2(\mathcal{O}_{\mathbb{C}_p}/p^r)$ is surjective. $\square$

Proposition 2.6. *One has*

$$\pi_1 K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \cong \mathcal{O}_{\mathbb{C}_p}^\times, \qquad \pi_0 K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \cong \mathbb{Z}, \qquad \pi_n K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) = 0 \quad (n < 0).$$

Proof. For a countable tower of spectra, the Milnor exact sequence [4, Theorem IX.3.1] reads

$$0 \longrightarrow \lim\nolimits_r^1 K_{n+1}(\mathcal{O}_{\mathbb{C}_p}/p^r) \longrightarrow \pi_n K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow \lim_r K_n(\mathcal{O}_{\mathbb{C}_p}/p^r) \longrightarrow 0. \tag{7}$$

The surjective K_2-tower of Lemma 2.5 is Mittag–Leffler, so its $\lim^1$ vanishes. Taking $n = 1$ in (7) and using the same lemma gives $\pi_1 K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) = \mathcal{O}_{\mathbb{C}_p}^\times$. Taking $n = 0$, the $\lim^1 K_1$ term vanishes and the constant K_0-tower of Lemma 2.4 has limit $\mathbb{Z}$. Finally, that lemma and the vanishing of $\lim^1 K_0(\mathcal{O}_{\mathbb{C}_p}/p^r)$ give $\pi_n K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) = 0$ for every $n < 0$. $\square$

Proposition 2.7. *The remaining homotopy groups of $K(\mathrm{Nuc}(\mathbb{C}_p))$ are*

$$K_1^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{C}_p^\times, \qquad K_0^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{Z}, \qquad K_n^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = 0 \quad (n < 0).$$

Proof. The field $\mathbb{C}_p$ has no negative K-groups. The valuation ring $\mathcal{O}_{\mathbb{C}_p}$ has no negative K-groups by [10, Theorem 1.3(iii)]. Together with Proposition 2.6, the cofiber sequence

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \longrightarrow K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathrm{Nuc}(\mathbb{C}_p))$$

of Proposition 2.2 shows first that $K_n^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = 0$ for $n < 0$.

In degrees zero and one, that cofiber sequence is obtained from

$$K(\mathcal{O}_{\mathbb{C}_p}) \xrightarrow{(i,-c)} K(\mathbb{C}_p) \oplus K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathrm{Nuc}(\mathbb{C}_p)),$$

where i is localization and c is completion. On K_0, the first map is $n \mapsto (n, -n)$ from $\mathbb{Z}$ to $\mathbb{Z} \oplus \mathbb{Z}$. It is injective and has cokernel $\mathbb{Z}$, so $K_0^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = \mathbb{Z}$.

On K_1, Proposition 2.6 identifies the map with

$$\mathcal{O}_{\mathbb{C}_p}^\times \longrightarrow \mathbb{C}_p^\times \times \mathcal{O}_{\mathbb{C}_p}^\times, \qquad u \longmapsto (u, u^{-1}).$$

The following boundary to $K_0(\mathcal{O}_{\mathbb{C}_p})$ is zero because the K_0 map just displayed is injective. Hence $K_1^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))$ is the cokernel of this graph map. The homomorphism $(x, v) \mapsto xv$ identifies that cokernel with $\mathbb{C}_p^\times$. $\square$

3. Finite and rational comparison

Finite coefficients determine the torsion and divisibility of the desired groups. The rational calculation rules out the remaining positive-even summands, and the support comparison transfers the formal calculation to $K(\mathrm{Nuc}(\mathbb{C}_p))$.

3.1. **Finite coefficients.**

Proposition 3.1. *For every integer $m \geq 2$, the canonical map*

$$K(\mathcal{O}_{\mathbb{C}_p})/m \longrightarrow K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/m$$

is an equivalence.

Proof. First take $m = q^a$ with q prime. If $q = p$, the pair $(\mathcal{O}_{\mathbb{C}_p}, p\mathcal{O}_{\mathbb{C}_p})$ is henselian because $\mathcal{O}_{\mathbb{C}_p}$ is p-adically complete; moreover, $\mathcal{O}_{\mathbb{C}_p}$ has bounded p-power torsion. The p-adic K-continuity theorem [5, Theorem 5.23] gives

$$K(\mathcal{O}_{\mathbb{C}_p})/p \xrightarrow{\sim} K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/p.$$

A mod-p equivalence is a mod-p^a equivalence for every $a \geq 1$, as follows inductively from the cofiber sequences between Moore spectra.

Now let $q = \ell \neq p$. Each $(\mathcal{O}_{\mathbb{C}_p}, p^r\mathcal{O}_{\mathbb{C}_p})$ is a henselian pair and ℓ is a unit in $\mathcal{O}_{\mathbb{C}_p}$. Gabber rigidity, in the form recorded in [5, Theorem 1.4], gives compatible equivalences

$$K(\mathcal{O}_{\mathbb{C}_p})/\ell^a \xrightarrow{\sim} K(\mathcal{O}_{\mathbb{C}_p}/p^r)/\ell^a \qquad (r \geq 1).$$

The Moore spectrum $\mathbb{S}/\ell^a$ is finite and hence dualizable. Smashing with it commutes with homotopy inverse limits, so passage to holim_r proves the claim for ℓ^a. On the fiber of $K(\mathcal{O}_{\mathbb{C}_p}) \to K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$, the prime-power cases say that multiplication by each q^{a_q} is an equivalence. If $m = \prod_q q^{a_q}$, multiplication by m is their composite and is therefore an equivalence on the fiber. The fiber modulo m vanishes, which proves the result for arbitrary m.

The cited continuity and rigidity theorems are stated for connective K-theory. The canonical maps from connective to nonconnective K-theory are equivalences for $\mathcal{O}_{\mathbb{C}_p}$ and for every $\mathcal{O}_{\mathbb{C}_p}/p^r$, by valuation-ring negative K-vanishing and Lemma 2.4. Proposition 2.6 gives the corresponding assertion for $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$. Thus the connective equivalences just proved give the asserted equivalence for our convention on K. □

Corollary 3.2. *For every $m \geq 2$ there is an equivalence*

$$K(\mathrm{Nuc}(\mathbb{C}_p))/m \simeq K(\mathbb{C}_p)/m.$$

Consequently, for every $i \geq 0$,

$$\pi_{2i}(K(\mathrm{Nuc}(\mathbb{C}_p))/m) \cong \mathbb{Z}/m(i), \qquad \pi_{2i+1}(K(\mathrm{Nuc}(\mathbb{C}_p))/m) = 0, \tag{8}$$

and $\pi_n(K(\mathrm{Nuc}(\mathbb{C}_p))/m) = 0$ for $n < 0$.

Proof. Smash the pushout of Proposition 2.2 with $\mathbb{S}/m$ and apply Proposition 3.1. This gives $K(\mathrm{Nuc}(\mathbb{C}_p))/m \simeq K(\mathbb{C}_p)/m$. Since $\mathbb{C}_p$ is algebraically closed of characteristic zero, Suslin rigidity and the finite-coefficient Bott calculation give

$$\pi_j(K(\mathbb{C}_p)/m) \cong \begin{cases} \mathbb{Z}/m(i), & j = 2i \geq 0, \\ 0, & j \text{ is odd or } j < 0. \end{cases}$$

See [14] or [15, Chapter VI, Theorem 1.3 and Proposition 1.4]. □

Remark 3.3. The groups in (8) have the connective ku/m pattern, not the periodic KU/m pattern. Finite coefficients do not determine the integral spectrum because they do not detect uniquely divisible summands.

3.2. **The rational formal calculation.**

Proposition 3.4. *The rational homotopy groups of $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$ are*

$$\pi_n(K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{\mathbb{Q}}) \cong \begin{cases} \mathbb{Q}, & n = 0, \\ B^+_{\mathrm{dR}}/\operatorname{Fil}^i B^+_{\mathrm{dR}}, & n = 2i-1 \geq 1, \\ 0, & n > 0 \textit{ is even}, \\ 0, & n < 0. \end{cases}$$

Proof. Theorem 4.14 and equation (29) of Antieau–Mathew–Morrow–Nikolaus [1] apply to every qcqs p-adic formal scheme with bounded p-power torsion. The affine formal scheme $\operatorname{Spf}(\mathcal{O}_{\mathbb{C}_p})$ satisfies these hypotheses because $\mathcal{O}_{\mathbb{C}_p}$ is p-torsion-free. Their theorem gives a cofiber sequence

$$K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{\mathbb{Q}} \longrightarrow K(\mathcal{O}_{\mathbb{C}_p}/p)_{\mathbb{Q}} \longrightarrow \bigoplus_{i\geq 0} \Sigma^{2i} H\Big((L\Omega_{\mathcal{O}_{\mathbb{C}_p}}/L\Omega^{\geq i}_{\mathcal{O}_{\mathbb{C}_p}})_{\mathbb{Q}_p}\Big). \tag{9}$$

Here $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{\mathbb{Q}}$ means the rationalization of $\operatorname{holim}_{r\geq 1} K(\mathcal{O}_{\mathbb{C}_p}/p^r)$.

Reduction modulo p identifies

$$\mathcal{O}_{\mathbb{C}_p}/p \cong \mathcal{O}_{\overline{\mathbb{Q}}_p}/p \cong \operatorname*{colim}_{L/\mathbb{Q}_p \text{ finite}} \mathcal{O}_L/p. \tag{10}$$

Each $\mathcal{O}_L/p$ is a finite local ring. It has $K_0 = \mathbb{Z}$, whereas its positive K-groups are finite [15, Chapter IV, Proposition 1.16] and its negative K-groups vanish. Perfect complexes descend through filtered colimits of rings, and nonconnective K-theory preserves the resulting filtered colimits [2, Definition 9.6, Lemma 9.7, and Theorem 9.8]. Hence

$$K(\mathcal{O}_{\mathbb{C}_p}/p)_{\mathbb{Q}} \simeq H\mathbb{Q}. \tag{11}$$

The transition maps in this special-fiber colimit induce the identity on K_0.

The ring $\mathcal{O}_{\mathbb{C}_p}$ is perfectoid. Construction 7.6 of [1] identifies $(L\Omega_{\mathcal{O}_{\mathbb{C}_p}})_{\mathbb{Q}_p}$ with B^+_{cris} and its Hodge completion with B^+_{dR}. Hodge completion does not change a finite Hodge quotient. Hence, for $i \geq 1$,

$$(L\Omega_{\mathcal{O}_{\mathbb{C}_p}}/L\Omega^{\geq i}_{\mathcal{O}_{\mathbb{C}_p}})_{\mathbb{Q}_p} \simeq B^+_{\mathrm{dR}}/\operatorname{Fil}^i B^+_{\mathrm{dR}}, \tag{12}$$

concentrated in homological degree zero; the $i = 0$ quotient is zero.

Substitution of (11) and (12) into (9) gives

$$K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{\mathbb{Q}} \longrightarrow H\mathbb{Q} \longrightarrow \bigoplus_{i\geq 1} \Sigma^{2i} H(B^+_{\mathrm{dR}}/\operatorname{Fil}^i B^+_{\mathrm{dR}}).$$

The long exact homotopy sequence gives the stated groups. In each positive degree at most one summand on the right contributes, so there is no product or convergence ambiguity. The negative-degree assertion is Proposition 2.6 after rationalization. □

3.3. **The support term.** The passage from the formal term $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$ to $K\big(\mathrm{Nuc}(\mathbb{C}_p)\big)$ is controlled by $K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p)$. We prove the integral equivalence $K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \simeq H\mathbb{Q}$. Its rational and derived p-complete consequences then follow immediately. Since $\mathcal{O}_{\mathbb{C}_p}$ is not a discrete valuation ring, the first step is descent to finite extensions of $\mathbb{Q}_p$.

Lemma 3.5. *Base change induces an equivalence*

$$\operatorname{Perf}(\mathcal{O}_{\overline{\mathbb{Q}}_p} \textit{ on } (p)) \xrightarrow{\sim} \operatorname{Perf}(\mathcal{O}_{\mathbb{C}_p} \textit{ on } (p)).$$

Consequently,

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \simeq \operatorname*{colim}_{L/\mathbb{Q}_p \textit{ finite}} K(k_L), \tag{13}$$

where k_L is the residue field of L.

Proof. Reduction modulo p is unchanged by p-adic completion, so $\mathcal{O}_{\overline{\mathbb{Q}}_p}/p \xrightarrow{\sim} \mathcal{O}_{\mathbb{C}_p}/p$. For $A = \mathcal{O}_{\overline{\mathbb{Q}}_p}$ or $\mathcal{O}_{\mathbb{C}_p}$, the element p is a non-zero-divisor and the category of perfect complexes supported on $V(p)$ is the thick subcategory generated by the Koszul complex

$$K_A(p) = [A \xrightarrow{p} A] \simeq A/p.$$

The derived endomorphism algebra of this generator has cohomology

$$H^* R\mathrm{Hom}_A(A/p, A/p) \cong (A/p) \oplus (A/p)[-1].$$

The base-change map between the two derived endomorphism algebras is a quasi-isomorphism because it induces $\mathcal{O}_{\overline{\mathbb{Q}}_p}/p \cong \mathcal{O}_{\mathbb{C}_p}/p$ on both cohomology groups. Morita theory for an idempotent-complete stable category with a compact generator gives the first equivalence.

Now $\mathcal{O}_{\overline{\mathbb{Q}}_p} = \mathrm{colim}_L \mathcal{O}_L$. Objects, morphisms, finite cones, and idempotents in the thick subcategory generated by a perfect complex descend to a finite stage. Taking the filtered colimit of these finite-stage thick subcategories therefore gives an equivalence

$$\mathrm{Perf}(\mathcal{O}_{\overline{\mathbb{Q}}_p} \text{ on } (p)) \simeq \operatorname*{colim}_L \mathrm{Perf}(\mathcal{O}_L \text{ on } (p)).$$

Nonconnective K-theory preserves filtered colimits by [2, Definition 9.6, Lemma 9.7, and Theorem 9.8]. Devissage for the discrete valuation ring $\mathcal{O}_L$ identifies $K(\mathcal{O}_L \text{ on } (p))$ with $K(k_L)$, proving (13). □

Lemma 3.6. *There is an equivalence*

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \simeq H\mathbb{Q}.$$

With the natural abstract $G_{\mathbb{Q}_p}$-action, this is an equivalence in $\mathrm{Fun}(BG^\delta_{\mathbb{Q}_p}, \mathrm{Sp})$, where $G_{\mathbb{Q}_p}$ acts trivially on $\mathbb{Q}$.

Proof. Use the colimit description (13). For $L \subset L'$, the $\mathcal{O}_{L'}$-module $\mathcal{O}_{L'} \otimes_{\mathcal{O}_L} k_L$ has a filtration of length $e(L'/L)$ whose graded pieces are $k_{L'}$. Devissage and additivity therefore identify the transition with

$$e(L'/L) \cdot \mathrm{res}^{k_{L'}}_{k_L} : K(k_L) \longrightarrow K(k_{L'}).$$

On K_0, residue-field extension preserves rank, so this map is multiplication by $e(L'/L)$. For every $d \geq 1$, adjoining a root of the Eisenstein polynomial $T^d - \pi_L$ gives a totally ramified extension of degree d. The compatible assignments

$$[n \text{ at the } L\text{-stage}] \longmapsto \frac{n}{e(L/\mathbb{Q}_p)}$$

identify the colimit of the copies of $\mathbb{Z}$ with $\mathbb{Q}$.

Now let $j > 0$ and let $x \in K_j(k_L)$. Quillen's calculation [12, Theorem 8] shows that x has finite order, say d. Choose a totally ramified extension L'/L of degree d. Its residue field is unchanged, so the preceding transition is multiplication by d and kills x. Thus every positive homotopy group of the colimit vanishes; the negative groups vanish at every stage. The support spectrum is connective and has only $\pi_0 = \mathbb{Q}$, which proves the equivalence.

The Galois action sends the rank generator at the L-stage to the rank generator at the gL-stage, so it acts trivially on the colimit $\mathbb{Q}$. The pointwise t-structure on $\mathrm{Fun}(BG^\delta_{\mathbb{Q}_p}, \mathrm{Sp})$ has heart $\mathrm{Fun}(BG^\delta_{\mathbb{Q}_p}, \mathbf{Ab})$ and gives the equivariant assertion. □

Corollary 3.7. *For every $a \geq 1$, one has $K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p)/p^a \simeq 0$ and $K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p)^\wedge_p \simeq 0$, also with the abstract Galois action.*

Proof. Multiplication by p^a is an equivalence on $H\mathbb{Q}$, so its cofiber vanishes. Taking the homotopy inverse limit over a gives the completion statement. □

Proposition 3.8. *For every $n \geq 2$, the map $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \to K(\mathrm{Nuc}(\mathbb{C}_p))$ induces an isomorphism*

$$\pi_n(K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{\mathbb{Q}}) \xrightarrow{\sim} K_n^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

In particular,

$$K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \otimes_{\mathbb{Z}} \mathbb{Q} = 0 \qquad (i \geq 1), \tag{14}$$

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \otimes_{\mathbb{Z}} \mathbb{Q} \cong B_{\mathrm{dR}}^{+}/\operatorname{Fil}^k B_{\mathrm{dR}}^{+} \qquad (k \geq 2). \tag{15}$$

Proof. Lemma 3.6 gives $K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p)_{\mathbb{Q}} \simeq H\mathbb{Q}$. The cofiber sequence

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \longrightarrow K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathrm{Nuc}(\mathbb{C}_p))$$

from (5) gives the asserted isomorphism in degrees at least two. Proposition 3.4 gives the formulas. ☐

4. Integral homotopy groups

For every integer $m \geq 2$, the cofiber sequence

$$K(\mathrm{Nuc}(\mathbb{C}_p)) \xrightarrow{m} K(\mathrm{Nuc}(\mathbb{C}_p)) \longrightarrow K(\mathrm{Nuc}(\mathbb{C}_p))/m$$

gives the natural short exact sequence

$$0 \longrightarrow K_n^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))/mK_n^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \longrightarrow \pi_n(K(\mathrm{Nuc}(\mathbb{C}_p))/m) \xrightarrow{\partial_m} K_{n-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))[m] \longrightarrow 0. \tag{16}$$

The two end terms are respectively the cokernel and kernel of multiplication by m on the indicated integral homotopy groups.

Proposition 4.1. *For every $i \geq 1$,*

$$K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = 0.$$

Proof. Apply (16) with $n = 2i+1$. Corollary 3.2 gives $\pi_{2i+1}(K(\mathrm{Nuc}(\mathbb{C}_p))/m) = 0$, and therefore $K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))[m] = 0$ for every $m \geq 2$. Thus $K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))$ is torsion-free. On the other hand, Proposition 3.8 gives

$$K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \otimes_{\mathbb{Z}} \mathbb{Q} \cong \pi_{2i}(K(\mathrm{Nuc}(\mathbb{C}_p))_{\mathbb{Q}}) = 0.$$

A torsion-free abelian group injects into its rationalization, so $K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = 0$. ☐

Remark 4.2 (The degree-two Bockstein). Finite coefficients make $K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))$ torsion-free; the rational calculation is needed to make it zero. In degree two, (16) therefore identifies

$$\pi_2(K(\mathrm{Nuc}(\mathbb{C}_p))/m) \xrightarrow{\sim} K_1^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))[m] = \mathbb{C}_p^{\times}[m] = \mu_m(\mathbb{C}_p).$$

Thus the class in $\pi_2(K(\mathrm{Nuc}(\mathbb{C}_p))/m)$ comes from K_1-torsion, not from an integral class in degree two.

Proposition 4.3. *Let $k \geq 2$. The group $K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))$ is divisible, and*

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))[m] \cong \mathbb{Z}/m(k) \qquad (m \geq 2), \tag{17}$$

$$\operatorname{tors}\Big(K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))\Big) \cong \bigoplus_{\ell} (\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k), \tag{18}$$

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))/\operatorname{tors}\Big(K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))\Big) \cong B_{\mathrm{dR}}^{+}/\operatorname{Fil}^k B_{\mathrm{dR}}^{+}. \tag{19}$$

Consequently, there is a noncanonical isomorphism of ordinary abelian groups

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \left(\bigoplus_{\ell}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\right) \oplus B_{\mathrm{dR}}^{+}/\operatorname{Fil}^k B_{\mathrm{dR}}^{+}. \tag{20}$$

Proof. Take $n = 2k-1$ in (16). The middle group is zero by Corollary 3.2; hence multiplication by every positive integer is surjective on $K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))$. This is precisely divisibility.

Next take $n = 2k$. Proposition 4.1 and (8) identify the Bockstein boundary with an isomorphism

$$\partial_m : \mathbb{Z}/m(k) \xrightarrow{\sim} K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))[m].$$

For a fixed prime ℓ, the ℓ-primary subgroup of the torsion subgroup of $K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))$ is a divisible ℓ-primary group. It is therefore isomorphic to $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)^{(\kappa_\ell)}$ for some cardinal κ_ℓ. Its subgroup killed by ℓ is $\mathbb{Z}/\ell(k)$ by the preceding Bockstein isomorphism, so $\kappa_\ell = 1$. Thus its underlying abelian group is $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. The compatible Galois-module identification is proved in Theorem 1.2. Every torsion element has finite order and hence only finitely many nonzero primary components.

For a divisible abelian group, localization to $\mathbb{Q}$ is surjective and has its torsion subgroup as kernel. Applying this observation to $K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p))$ and using (15) proves (19). Finally, every divisible abelian group is injective in **Ab**. Its divisible torsion subgroup therefore splits off, which proves (20). This argument does not select a splitting. □

Proposition 4.4 (Crystalline form)**.** *For every $k \geq 2$, there is a noncanonical isomorphism in* **Ab**

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \left(\bigoplus_{\ell\neq p}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\right) \oplus \frac{(B_{\mathrm{cris}}^{+})^{\varphi=p^k}}{\mathbb{Z}_p(k)}. \tag{21}$$

Proof. Fontaine's fundamental exact sequence, in the form proved in [1, Theorem 7.7 and Example 7.8], is

$$0 \longrightarrow \mathbb{Q}_p(k) \longrightarrow (B_{\mathrm{cris}}^{+})^{\varphi=p^k} \longrightarrow B_{\mathrm{dR}}^{+}/\operatorname{Fil}^k B_{\mathrm{dR}}^{+} \longrightarrow 0. \tag{22}$$

Via the first injection, regard

$$\mathbb{Z}_p(k) \subset \mathbb{Q}_p(k) \subset (B_{\mathrm{cris}}^{+})^{\varphi=p^k}$$

as the usual Tate lattice. Quotienting the first two terms by this lattice gives

$$0 \longrightarrow (\mathbb{Q}_p/\mathbb{Z}_p)(k) \longrightarrow \frac{(B_{\mathrm{cris}}^{+})^{\varphi=p^k}}{\mathbb{Z}_p(k)} \longrightarrow B_{\mathrm{dR}}^{+}/\operatorname{Fil}^k B_{\mathrm{dR}}^{+} \longrightarrow 0. \tag{23}$$

After forgetting the Galois action and topology, the left-hand group is divisible and hence injective. Sequence (23) therefore splits in **Ab**, though not naturally. Replacing the summand $(\mathbb{Q}_p/\mathbb{Z}_p)(k) \oplus B_{\mathrm{dR}}^{+}/\operatorname{Fil}^k B_{\mathrm{dR}}^{+}$ in (20) by the isomorphic middle term of (23) proves (21). □

Proof of Theorem 1.1. Proposition 2.7 gives

$$K_0^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{Z}, \qquad K_1^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \mathbb{C}_p^{\times}, \qquad K_n^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = 0 \quad (n<0).$$

Proposition 4.1 gives $K_{2i}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) = 0$ for every $i \geq 1$. Finally, for $k \geq 2$, Proposition 4.4 gives

$$K_{2k-1}^{\mathrm{cont}}(\mathrm{Nuc}(\mathbb{C}_p)) \cong \left(\bigoplus_{\ell\neq p}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\right) \oplus \frac{(B_{\mathrm{cris}}^{+})^{\varphi=p^k}}{\mathbb{Z}_p(k)}.$$

These cases exhaust all integers. □

5. Galois-equivariant reconstruction

The action of $G_{\mathbb{Q}_p}$ on $\mathbb{C}_p$ induces, by scalar twisting, exact autoequivalences of $\mathrm{Nuc}(\mathbb{C}_p)$ and hence a homotopy-coherent action on $K(\mathrm{Nuc}(\mathbb{C}_p))$. We write $G^{\delta}_{\mathbb{Q}_p}$ when the profinite topology is forgotten. Thus the action considered in this section is an object of

$$\mathrm{Fun}(BG^{\delta}_{\mathbb{Q}_p}, \mathrm{Sp}),$$

and its homotopy groups are abstract $G_{\mathbb{Q}_p}$-modules. This category is strong enough to retain the Fontaine extension class, but it does not require the action on a homotopy group with the discrete topology to be continuous.

The ordinary proof of Proposition 4.4 forgets this action before splitting a divisible extension. We instead identify the p-local extension from the cyclotomic-trace square of [1, (23)], then split the prime-to-p torsion by reduction to $\mathcal{O}_{\mathbb{C}_p}/p$. These are the two inputs in the proof of Theorem 1.2.

5.1. **Equivariant endpoints.**

Proposition 5.1 (Equivariant end terms)**.** *For every $m \geq 2$, the finite-coefficient calculation is $G^{\delta}_{\mathbb{Q}_p}$-equivariant:*

$$\pi_j(K(\mathrm{Nuc}(\mathbb{C}_p))/m) \cong \begin{cases} \mathbb{Z}/m(i), & j = 2i \geq 0, \\ 0, & j \text{ is odd or } j < 0. \end{cases}$$

Here the Tate twist has the natural Galois action. For every $k \geq 2$, rationalization gives an isomorphism

$$K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p)) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\sim} B^{+}_{\mathrm{dR}}/\operatorname{Fil}^k B^{+}_{\mathrm{dR}}$$

of abstract $G_{\mathbb{Q}_p}$-modules.

Proof. Every map in the formal/Tate square of Proposition 2.2 is natural in the pair $\mathcal{O}_{\mathbb{C}_p} \subset \mathbb{C}_p$. Applying this square to the automorphisms supplied by $G_{\mathbb{Q}_p}$ therefore places it in $\mathrm{Fun}(BG^{\delta}_{\mathbb{Q}_p}, \mathrm{Sp})$. The finite-coefficient continuity maps used in Proposition 3.1, Gabber rigidity, and the Suslin–Bott calculation are natural in the ring. Under the chosen Kummer–Bott normalization, Suslin's even-degree class is $\mu_m(\mathbb{C}_p)^{\otimes i}$; see [15, Chapter VI, Proposition 1.7.1]. This proves the first assertion, including its Galois action.

The rational Beilinson comparison used in Proposition 3.4 is likewise natural. In the perfectoid case, the identification of the finite Hodge quotient with $B^{+}_{\mathrm{dR}}/\operatorname{Fil}^k B^{+}_{\mathrm{dR}}$ is $G_{\mathbb{Q}_p}$-equivariant. The support term is rationally concentrated in degree zero by Proposition 3.8; hence the formal-to-Tate map transports this identification to $K(\mathrm{Nuc}(\mathbb{C}_p))$ in degree $2k-1 \geq 3$. □

5.2. **The arithmetic gluing map.**

Proposition 5.2 (The arithmetic gluing morphism)**.** *For every $k \geq 2$, the component in degrees $2k-1$ and $2k$ of the arithmetic comparison*

$$(K(\mathrm{Nuc}(\mathbb{C}_p)))_{\mathbb{Q}} \longrightarrow ((K(\mathrm{Nuc}(\mathbb{C}_p)))^{\wedge}_p)[1/p]$$

is, in $\mathrm{Fun}(BG^{\delta}_{\mathbb{Q}_p}, \mathrm{Sp})$, the map

$$\Sigma^{2k-1} H\Big(B^{+}_{\mathrm{dR}}/\operatorname{Fil}^k B^{+}_{\mathrm{dR}}\Big) \longrightarrow \Sigma^{2k} H\mathbb{Q}_p(k) \tag{24}$$

represented by $\lambda_k \partial_k$ for some $\lambda_k \in \mathbb{Q}_p^{\times}$. Here ∂_k is the connecting class of Fontaine's fundamental exact sequence (22), now regarded as a $G_{\mathbb{Q}_p}$-equivariant sequence. With the endpoint coordinates furnished by AMMN, the map is Fontaine's boundary; λ_k records only the comparison with the Kummer–Bott coordinate. Only the nonvanishing of λ_k is used below.

Proof. Corollary 3.7 and Proposition 3.8, applied to the natural cofiber sequence

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p};p) \longrightarrow K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathrm{Nuc}(\mathbb{C}_p)),$$

show that the last map identifies p-completions and rational homotopy groups in degrees at least two. Naturality also identifies the arithmetic comparison maps in the two degrees under consideration. It is therefore enough to calculate the comparison for $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$.

The inverse-limit Dundas–Goodwillie–McCarthy square of [1, (23) and Proposition 4.2] is the natural cartesian square

$$\begin{tikzcd} K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \arrow[r] \arrow[d, "\mathrm{tr}^{\mathrm{cts}}"'] & K(\mathcal{O}_{\mathbb{C}_p}/p) \arrow[d, "\mathrm{tr}"] \\ \mathrm{TC}(\mathcal{O}_{\mathbb{C}_p};\mathbb{Z}_p) \arrow[r] & \mathrm{TC}(\mathcal{O}_{\mathbb{C}_p}/p;\mathbb{Z}_p). \end{tikzcd} \tag{25}$$

It is a square in $\mathrm{Fun}(BG^{\delta}_{\mathbb{Q}_p},\mathrm{Sp})$ because the cyclotomic trace and all four constructions are natural in $\mathcal{O}_{\mathbb{C}_p}$. CMM's p-adic K-continuity theorem [5, Theorem 5.23] identifies the p-completion of $K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$ with that of $K(\mathcal{O}_{\mathbb{C}_p})$. Since $\mathcal{O}_{\mathbb{C}_p}$ is strictly henselian local of residue characteristic p, the cyclotomic trace then identifies the latter with the connective part of $\mathrm{TC}(\mathcal{O}_{\mathbb{C}_p};\mathbb{Z}_p)$ [5, Theorem 6.1 and Corollary 6.9]. Together with the TC-continuity used in (25), this makes the connective p-completion of the left vertical map an equivalence. This is sufficient in degrees $2k-1, 2k > 0$.

Moreover, $K(\mathcal{O}_{\mathbb{C}_p}/p)_{\mathbb{Q}} \simeq H\mathbb{Q}$ by (11). Thus, in positive degrees, the rational formal term is the fiber of the bottom horizontal map in (25), and the arithmetic comparison is the corresponding fiber inclusion into $\mathrm{TC}(\mathcal{O}_{\mathbb{C}_p};\mathbb{Q}_p)$.

To pass from the associated-graded square to the actual TC morphism, we use the functorial rational Adams splitting. By [1, Corollary 6.23 and its proof], the complete motivic filtration on $\mathrm{TC}(A;\mathbb{Q}_p)$ has graded pieces $\mathbb{Q}_p(i)(A)[2i]$ and splits functorially in the simplicial commutative ring A. Applied to $\mathcal{O}_{\mathbb{C}_p} \to \mathcal{O}_{\mathbb{C}_p}/p$, this identifies the weight-k summand of the bottom horizontal map in (25) with

$$\mathbb{Q}_p(k)(\mathcal{O}_{\mathbb{C}_p})[2k] \longrightarrow \mathbb{Q}_p(k)(\mathcal{O}_{\mathbb{C}_p}/p)[2k].$$

Theorem 6.17 of [1] identifies the fiber of this map with the weight-k Beilinson fiber. All these filtrations, splittings, and comparison maps are natural in A. Hence, after composition with the action functor $BG^{\delta}_{\mathbb{Q}_p} \to \mathrm{Fun}(\Delta^1,\mathrm{CAlg})$ determined by the equivariant reduction map $\mathcal{O}_{\mathbb{C}_p} \to \mathcal{O}_{\mathbb{C}_p}/p$, the identification holds in $\mathrm{Fun}(BG^{\delta}_{\mathbb{Q}_p}, D(\mathbb{Q}_p))$.

For $A = \mathcal{O}_{\mathbb{C}_p}$, Theorem 7.7 and Example 7.8 of [1] identify this weight-k fiber square with Fontaine's fundamental square. Its fiber is $B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}[-1]$, so after the shift $[2k]$ the fiber inclusion is

$$\Sigma^{2k-1} H\Big(B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}\Big) \longrightarrow \Sigma^{2k} H\mathbb{Q}_p(k),$$

the connecting morphism ∂_k of (22). Remark 7.9 of [1] identifies, for $\mathbb{C}_p$, the period map in weight one with the Galois-equivariant Chern character $d\log$; multiplicativity and generation in degree one determine the higher-weight maps. Thus the AMMN endpoint coordinates give Fontaine's boundary. Comparing the $\mathbb{Q}_p(k)$ endpoint with the Kummer–Bott coordinate coming from p-completed K-theory changes it by an automorphism of a one-dimensional $\mathbb{Q}_p$-vector space. Since the cyclotomic trace is an equivalence in the degrees under consideration, this automorphism is multiplication by some $\lambda_k \in \mathbb{Q}_p^{\times}$. Consequently the arithmetic gluing morphism in the coordinates used here is $\lambda_k \partial_k$. □

5.3. **The p-primary Fontaine extension.**

Theorem 5.3 (The p-primary Fontaine extension)**.** *For every $k \geq 2$, there is a natural short exact sequence in $\mathbf{Ab}^{G^{\delta}_{\mathbb{Q}_p}}$*

$$0 \longrightarrow (\mathbb{Q}_p/\mathbb{Z}_p)(k) \longrightarrow \pi_{2k-1}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))_{(p)}\big) \longrightarrow B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}} \longrightarrow 0. \tag{26}$$

Its middle term has the Galois-equivariant isomorphism type

$$\pi_{2k-1}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))_{(p)}\big) \cong \frac{(B^+_{\mathrm{cris}})^{\varphi=p^k}}{\mathbb{Z}_p(k)} \tag{27}$$

as an abstract $G_{\mathbb{Q}_p}$-module. The isomorphism in (27) *is defined only up to a nonzero scalar rescaling of the quotient $B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}$.*

Proof. The negative-degree calculation in Section 2.1 shows that $K(\mathrm{Nuc}(\mathbb{C}_p))$ is connective. Its p-local arithmetic fracture square therefore gives the natural fiber sequence

$$\begin{aligned}(K(\mathrm{Nuc}(\mathbb{C}_p)))_{(p)} \longrightarrow &(K(\mathrm{Nuc}(\mathbb{C}_p)))^{\wedge}_p \oplus (K(\mathrm{Nuc}(\mathbb{C}_p)))_{\mathbb{Q}} \\ &\xrightarrow{\iota - c} ((K(\mathrm{Nuc}(\mathbb{C}_p)))^{\wedge}_p)[1/p].\end{aligned} \tag{28}$$

It is a fiber sequence in $\mathrm{Fun}(BG^{\delta}_{\mathbb{Q}_p}, \mathrm{Sp})$ because arithmetic localization and completion are functorial. Here ι is the localization of the completed term and c is the arithmetic comparison of Proposition 5.2; this fixes the sign convention for the extension class below.

The square is cartesian for the following elementary reason. The map from $(K(\mathrm{Nuc}(\mathbb{C}_p)))_{(p)}$ to the displayed homotopy pullback is both a rational equivalence and a mod-p equivalence; the latter uses that derived p-completion is a mod-p equivalence. Its cofiber is p-local. Vanishing modulo p makes multiplication by p an equivalence on that cofiber, so it is rational; its vanishing rationalization then makes it contractible.

Finite-coefficient rigidity and the vanishing of the adjacent odd groups give

$$\pi_{2k}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))^{\wedge}_p\big) = \mathbb{Z}_p(k), \qquad \pi_{2k-1}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))^{\wedge}_p\big) = 0.$$

The inverse system $\mathbb{Z}/p^r(k)$ has surjective transition maps, so there is no Milnor $\lim^1$ term in this assertion. After inverting p, the corresponding groups are $\mathbb{Q}_p(k)$ and zero. On the rational side, Proposition 5.1 gives

$$\pi_{2k-1}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))_{\mathbb{Q}}\big) = B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}, \qquad \pi_{2k}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))_{\mathbb{Q}}\big) = 0.$$

The homotopy long exact sequence of (28) therefore contains the following five-term segment, where the vanishing on the far left also uses Proposition 4.1:

$$0 \longrightarrow \mathbb{Z}_p(k) \longrightarrow \mathbb{Q}_p(k) \longrightarrow \pi_{2k-1}\big((K(\mathrm{Nuc}(\mathbb{C}_p)))_{(p)}\big) \longrightarrow B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}} \longrightarrow 0,$$

which yields (26).

It remains to identify its extension class. In degrees $2k-1$ and $2k$, the last arrow of (28) is

$$\Sigma^{2k} H\mathbb{Z}_p(k) \oplus \Sigma^{2k-1} H\Big(B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}\Big) \xrightarrow{(\iota, -\lambda_k \partial_k)} \Sigma^{2k} H\mathbb{Q}_p(k).$$

Apply the 3×3 lemma to the cofiber sequence $\mathbb{Z}_p(k) \to \mathbb{Q}_p(k) \to (\mathbb{Q}_p/\mathbb{Z}_p)(k)$ and to the extension represented by $\lambda_k \partial_k$. The two-stage homotopy fiber is the pushout of that extension along

$$\mathbb{Q}_p(k) \longrightarrow (\mathbb{Q}_p/\mathbb{Z}_p)(k).$$

The sign from $\iota - c$ can be absorbed into the nonzero scalar λ_k. By Proposition 5.2, the first extension is therefore a pullback of (22) along a nonzero scalar on $B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}$. Rescaling that quotient identifies its middle term with $(B^+_{\mathrm{cris}})^{\varphi=p^k}$ as an abstract $G_{\mathbb{Q}_p}$-module and is the identity on the kernel $\mathbb{Q}_p(k)$; hence it preserves the fixed lattice $\mathbb{Z}_p(k)$. The indicated pushout then gives $(B^+_{\mathrm{cris}})^{\varphi=p^k}/\mathbb{Z}_p(k)$ and proves the theorem. □

5.4. **Splitting the prime-to-p torsion.**

Proposition 5.4 (Special-fiber splitting)**.** *For every $k \geq 2$, reduction to $\mathcal{O}_{\mathbb{C}_p}/p$ induces a $G^\delta_{\mathbb{Q}_p}$-equivariant retraction*

$$K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p)) \longrightarrow \bigoplus_{\ell \neq p} K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p))[\ell^\infty].$$

Consequently, the localization sequence (1) *is split in* $\mathbf{Ab}^{G^\delta_{\mathbb{Q}_p}}$.

Proof. Lemma 3.6 and the cofiber sequence

$$K_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \longrightarrow K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathrm{Nuc}(\mathbb{C}_p))$$

give a $G^\delta_{\mathbb{Q}_p}$-equivariant isomorphism

$$\pi_j K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \xrightarrow{\sim} K^{\mathrm{cont}}_j(\mathrm{Nuc}(\mathbb{C}_p)) \qquad (j \geq 2). \tag{29}$$

Let

$$\rho : K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow K(\mathcal{O}_{\mathbb{C}_p}/p)$$

be projection from the homotopy inverse limit to its first stage.

Fix a prime $\ell \neq p$ and an integer $a \geq 1$. For every $r \geq 1$, the kernel of $\mathcal{O}_{\mathbb{C}_p}/p^r \to \mathcal{O}_{\mathbb{C}_p}/p$ is nilpotent. This is a henselian pair in which ℓ is invertible, so Gabber rigidity [5, Theorem 1.4] gives a natural equivalence

$$K(\mathcal{O}_{\mathbb{C}_p}/p^r)/\ell^a \xrightarrow{\sim} K(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a.$$

The cited rigidity theorem is formulated for connective K-theory. The negative groups of both rings vanish by Lemma 2.4, so the same equivalence holds for the nonconnective convention used here. The Moore spectrum $\mathbb{S}/\ell^a$ is finite and dualizable; smashing with it commutes with homotopy inverse limits. The displayed equivalences are compatible with reduction in r, and therefore

$$\rho/\ell^a : K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/\ell^a \xrightarrow{\sim} K(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a \tag{30}$$

is an equivalence in $\mathrm{Fun}(BG^\delta_{\mathbb{Q}_p}, \mathrm{Sp})$.

The equality

$$\mathcal{O}_{\mathbb{C}_p}/p = \operatorname*{colim}_{L/\mathbb{Q}_p \text{ finite}} \mathcal{O}_L/p$$

and preservation of filtered colimits by nonconnective K-theory show that $K_j(\mathcal{O}_{\mathbb{C}_p}/p)$ is torsion for $j > 0$: every $\mathcal{O}_L/p$ is a finite ring, whose positive K-groups are finite [15, Chapter IV, Proposition 1.16]. Moreover, Lemma 3.6, (30), and Proposition 5.1 give

$$K(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a \simeq K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/\ell^a \simeq K(\mathrm{Nuc}(\mathbb{C}_p))/\ell^a,$$

and hence

$$\pi_{2i+1}(K(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a) = 0 \qquad (i \geq 0).$$

We next compare the actual primary torsion subgroups. The Bockstein short exact sequence for $K(\mathcal{O}_{\mathbb{C}_p}/p)$ in degree $2k+1$ implies

$$K_{2k}(\mathcal{O}_{\mathbb{C}_p}/p)[\ell^a] = 0.$$

The group $K_{2k}(\mathcal{O}_{\mathbb{C}_p}/p)$ is torsion and has no ℓ^a-torsion. Hence every element has finite order prime to ℓ, and Bézout's identity supplies its unique ℓ^a-division. Multiplication by ℓ^a is therefore an automorphism. Thus

$$K_{2k}(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a = 0.$$

On the formal side, (29) and Proposition 4.1 give $\pi_{2k}K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) = 0$. The natural Bockstein square in degree $2k$ is consequently

$$\begin{tikzcd}
\pi_{2k}(K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/\ell^a) \arrow[r, "\partial"] \arrow[d, "\sim"'] & \pi_{2k-1}K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})[\ell^a] \arrow[d, "\rho_*"] \\
\pi_{2k}\big(K(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a\big) \arrow[r, "\partial"] & K_{2k-1}(\mathcal{O}_{\mathbb{C}_p}/p)[\ell^a].
\end{tikzcd}$$

Both horizontal arrows are isomorphisms, and the left vertical arrow is an isomorphism by (30). Hence ρ_* is an isomorphism on ℓ^a-torsion. Taking the union over a and then the direct sum over $\ell \neq p$ gives an actual equivariant isomorphism between the prime-to-p torsion subgroups of $\pi_{2k-1}K_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$ and $K_{2k-1}(\mathcal{O}_{\mathbb{C}_p}/p)$.

Write $T_{p'}(M) = \bigoplus_{\ell\neq p} M[\ell^\infty]$. Because $K_{2k-1}(\mathcal{O}_{\mathbb{C}_p}/p)$ is a torsion group, its primary decomposition supplies a canonical $G_{\mathbb{Q}_p}$-equivariant projection

$$\mathrm{pr}_{p'} : K_{2k-1}(\mathcal{O}_{\mathbb{C}_p}/p) \longrightarrow T_{p'}\big(K_{2k-1}(\mathcal{O}_{\mathbb{C}_p}/p)\big).$$

Transport ρ_* across (29), and denote the resulting map from $K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p))$ to $K_{2k-1}(\mathcal{O}_{\mathbb{C}_p}/p)$ by $\widetilde{\rho}_*$. Its restriction to prime-to-p torsion is the isomorphism

$$\widetilde{\rho}_{p'} : T_{p'}\big(K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p))\big) \xrightarrow{\sim} T_{p'}\big(K_{2k-1}(\mathcal{O}_{\mathbb{C}_p}/p)\big).$$

Therefore

$$r_k = \widetilde{\rho}_{p'}^{-1} \circ \mathrm{pr}_{p'} \circ \widetilde{\rho}_* : K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p)) \longrightarrow T_{p'}\big(K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p))\big)$$

is an equivariant retraction.

Finally, Proposition 4.3 shows that $K^{\mathrm{cont}}_{2k-1}(\mathrm{Nuc}(\mathbb{C}_p))$ is divisible. Its localization at p is therefore surjective, with kernel precisely its prime-to-p torsion. The map formed by r_k and localization identifies it equivariantly with the direct sum of that kernel and $\pi_{2k-1}((K(\mathrm{Nuc}(\mathbb{C}_p)))_{(p)})$. The compatible Bockstein isomorphisms of Proposition 5.1 identify the kernel with

$$\bigoplus_{\ell\neq p}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k).$$

Here the Moore map from modulus ℓ^a to modulus ℓ^{a+1} induces multiplication by ℓ on the coefficient copy of $\mathbb{Z}/\ell^a(k)$; naturality of the boundary identifies it with the inclusion of ℓ^a-torsion into ℓ^{a+1}-torsion. Explicitly, given a prime-to-p torsion element and a localized class, choose any lift of the localized class and correct it by the difference between the prescribed torsion element and its image under r_k. This proves surjectivity of the direct-sum map; its kernel is immediate from $r_k|_{T_{p'}} = \mathrm{id}$. This proves the split sequence. □

Proof of Theorem 1.2. The Milnor and localization sequences used in Proposition 2.7 are natural in $\mathcal{O}_{\mathbb{C}_p} \subset \mathbb{C}_p$. Thus the identifications in degrees zero and one are equivariant, and the vanishing groups carry their unique Galois action.

For $k \geq 2$, Proposition 5.4 gives the split sequence (1). The compatible Bockstein isomorphisms of Proposition 5.1 identify its kernel with

$$\bigoplus_{\ell\neq p}(\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k),$$

and Theorem 5.3 identifies the quotient with the crystalline term in (2). This proves the theorem. □

6. Condensed Galois modules

We prove Theorem 1.3 by retaining the dependence on every extremally disconnected profinite test object. The argument first identifies the condensed continuous K-theory model and removes its support term, then computes its finite, completed, and rational endpoints, and finally identifies the arithmetic gluing morphism before applying fracture.

6.1. **The condensed K-theory model.** Retain the cardinal κ, the functors Δ and $\underline{(-)}^{\mathrm{top}}$, and the Fontaine topologies fixed before Theorem 1.3. We denote the resulting strict fundamental sequence by

$$0 \longrightarrow \mathbb{Q}_p(k) \longrightarrow (B^+_{\mathrm{cris}})^{\varphi=p^k} \longrightarrow B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}} \longrightarrow 0. \tag{31}$$

Dahlhausen–Yaylali–Zhou prove an equivalence in condensed spectra

$$\underline{K}^{\mathrm{nuc}} \simeq \gamma_\kappa \underline{K}^{\mathrm{cont}}_{\mathrm{pro}} \tag{32}$$

[6, Proposition 4.11]. For an extremally disconnected profinite set S, their construction gives

$$\underline{K}^{\mathrm{nuc}}(\mathbb{C}_p)(S) \simeq K\big(\mathrm{Nuc}(\mathrm{Cont}(S, \mathbb{C}_p))\big)\,. \tag{33}$$

Their Proposition 4.13 supplies hyperdescent. The internal Galois coaction is induced, naturally in S, by

$$\mathrm{Cont}(S, \mathbb{C}_p) \longrightarrow \mathrm{Cont}(S \times G_{\mathbb{Q}_p}, \mathbb{C}_p), \qquad f \longmapsto \big((s,g) \longmapsto g(f(s))\big). \tag{34}$$

Define the formal condensed spectrum by

$$\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) := \varprojlim_{r\geq 1} \Delta K(\mathcal{O}_{\mathbb{C}_p}/p^r). \tag{35}$$

Nonconnective K-theory preserves filtered colimits and finite products. Since continuous functions from a profinite set to $\mathcal{O}_{\mathbb{C}_p}/p^r$ are locally constant, (35) satisfies

$$\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})(S) \simeq \varprojlim_{r\geq 1} K\big(\mathrm{Cont}(S, \mathcal{O}_{\mathbb{C}_p})/p^r\big)$$

for every extremally disconnected profinite S. The formal/Tate triangle has the condensed realization

$$\underline{K}_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \longrightarrow \underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow \underline{K}^{\mathrm{nuc}}(\mathbb{C}_p). \tag{36}$$

6.2. **Support and period objects on profinite families.**

Lemma 6.1. *There is a natural internally $\underline{G}_{\mathbb{Q}_p}$-equivariant equivalence $\underline{K}_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p) \simeq \Delta H\mathbb{Q}$. Consequently,*

$$\pi^{\mathrm{cond}}_j \underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \xrightarrow{\sim} \pi^{\mathrm{cond}}_j \underline{K}^{\mathrm{nuc}}(\mathbb{C}_p) \qquad (j \geq 2). \tag{37}$$

Proof. At the point, dévissage gives

$$K(\mathcal{O}_{\mathbb{C}_p} \text{ on } (p)) \simeq \operatorname*{colim}_{L/\mathbb{Q}_p \text{ finite}} K(k_L).$$

The transition for $L \subset L'$ is residue-field extension followed by multiplication by the ramification index. Totally ramified extensions of arbitrarily divisible degree kill every positive finite-field K-class; the degree-zero colimit is $\mathbb{Q}$, and negative K-groups vanish.

For profinite S, the category of perfect p-power-torsion $\mathrm{Cont}(S, \mathcal{O}_{\mathbb{C}_p})$-complexes is generated by $\mathrm{Cont}(S, \mathcal{O}_{\mathbb{C}_p})/p$, and base change from locally constant functions is a Morita equivalence on this supported category. The same dévissage calculation gives

$$\underline{K}_{\mathrm{supp}}(\mathcal{O}_{\mathbb{C}_p}; p)(S) \simeq H\,\mathrm{LC}(S, \mathbb{Q}),$$

the value of $\Delta H\mathbb{Q}$ on S. Equation (37) follows from (36). □

Lemma 6.2 (Period objects on profinite families)**.** *For every extremally disconnected profinite set S, naturally in S, one has topological isomorphisms*

$$B_{\mathrm{dR}}^{+}(\mathrm{Cont}(S,\mathcal{O}_{\mathbb{C}_p}))/\,\mathrm{Fil}^i \cong \mathrm{Cont}(S, B_{\mathrm{dR}}^{+}/\,\mathrm{Fil}^i\, B_{\mathrm{dR}}^{+}) \qquad (i \geq 1) \tag{38}$$

and

$$B_{\mathrm{cris}}^{+}(\mathrm{Cont}(S,\mathcal{O}_{\mathbb{C}_p}))^{\varphi=p^k} \cong \mathrm{Cont}(S, (B_{\mathrm{cris}}^{+})^{\varphi=p^k}) \qquad (k \geq 1). \tag{39}$$

Under these identifications the family period maps assemble to the strict short exact sequence

$$0 \longrightarrow \underline{\mathbb{Q}_p(k)}^{\mathrm{top}} \longrightarrow \underline{(B_{\mathrm{cris}}^{+})^{\varphi=p^k}}^{\mathrm{top}} \longrightarrow \underline{B_{\mathrm{dR}}^{+}/\,\mathrm{Fil}^k\, B_{\mathrm{dR}}^{+}}^{\mathrm{top}} \longrightarrow 0 \tag{40}$$

in condensed Galois modules.

Proof. The ring $\mathrm{Cont}(S,\mathcal{O}_{\mathbb{C}_p})$ is integral perfectoid. Scholze's product formula [13, Corollary 6.6], equivalently Bosco's family comparison [3, Corollary 4.9], gives (38) topologically and naturally in S.

The map

$$\mathrm{LC}(S,\mathcal{O}_{\mathbb{C}_p}) \longrightarrow \mathrm{Cont}(S,\mathcal{O}_{\mathbb{C}_p})$$

exhibits the target as the derived p-completion of the source. Derived p-completion and the p-complete left-Kan-extension construction of derived de Rham cohomology give

$$A_{\mathrm{cris}}(\mathrm{Cont}(S,\mathcal{O}_{\mathbb{C}_p})) \cong \mathrm{Cont}(S, A_{\mathrm{cris}}(\mathcal{O}_{\mathbb{C}_p})).$$

After inverting p, compactness of S supplies a uniform denominator, and Frobenius acts pointwise. Taking the equalizer of Frobenius and multiplication by p^k proves (39).

The comparisons in [3, Corollary 4.9, Proposition 4.16, and Lemma A.33] identify the degree-zero period maps with the maps obtained by applying $\mathrm{Cont}(S,-)$ to (31). Their strictness and naturality give (40). The same identifications are natural for the maps induced by (34), so the sequence carries its internal Galois action. ☐

Lemma 6.3 (Topology of the crystalline quotient)**.** *The quotient map induces a canonical internally $\underline{G_{\mathbb{Q}_p}}$-equivariant isomorphism*

$$\mathrm{coker}\Big(\underline{\mathbb{Z}_p(k)}^{\mathrm{top}} \longrightarrow \underline{(B_{\mathrm{cris}}^{+})^{\varphi=p^k}}^{\mathrm{top}}\Big) \cong \underline{(B_{\mathrm{cris}}^{+})^{\varphi=p^k}/\mathbb{Z}_p(k)}^{\mathrm{top}}. \tag{41}$$

Proof. Sequence (31) is strict, and its one-dimensional kernel is continuously complemented after the Galois action is forgotten. The quotient by the compact lattice $\mathbb{Z}_p(k)$ admits a continuous section as a map of spaces. Hence $\mathrm{Cont}(S,-)$ preserves this cokernel for every profinite S. Naturality for the Galois action assembles these identifications into (41). ☐

6.3. **Endpoint calculations.**

Proposition 6.4 (Finite, completed, and rational endpoints)**.** *For every $m \geq 2$, there is a natural equivalence*

$$\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/m \simeq \Delta(K(\mathbb{C}_p)/m), \tag{42}$$

and therefore

$$\pi_j^{\mathrm{cond}}(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/m) \cong \begin{cases} \Delta\mathbb{Z}/m(i), & j = 2i \geq 0, \\ 0, & j \text{ odd or } j < 0. \end{cases} \tag{43}$$

For $k \geq 2$, the p-complete and overlap terms satisfy

$$\pi_{2k}^{\mathrm{cond}}(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}))_p^{\wedge} \cong \underline{\mathbb{Z}_p(k)}^{\mathrm{top}}, \qquad \pi_{2k-1}^{\mathrm{cond}}(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}))_p^{\wedge} = 0, \tag{44}$$

$$\pi_{2k}^{\mathrm{cond}}(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_p^{\wedge}[1/p]) \cong \underline{\mathbb{Q}_p(k)}^{\mathrm{top}}. \tag{45}$$

The rational endpoint is

$$\pi_{2k-1}^{\mathrm{cond}}(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})\otimes\mathbb{Q}) \cong \underline{B_{\mathrm{dR}}^{+}/\operatorname{Fil}^{k} B_{\mathrm{dR}}^{+}}^{\mathrm{top}}, \qquad \pi_{2k}^{\mathrm{cond}}(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})\otimes\mathbb{Q}) = 0. \tag{46}$$

Finally,

$$\pi_{2i}^{\mathrm{cond}}\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) = 0 \qquad (i>0). \tag{47}$$

Proof. Finite coefficients. For $m=p$, [5, Theorem 5.23] gives the comparison after Postnikov, or protruncated, localization of the relevant pro-spectrum. In each fixed degree the homotopy tower of the fiber is pro-zero. The condensed Milnor sequence [6, Lemma 3.23] therefore kills both its inverse-limit and derived inverse-limit terms, which proves (42) for $m=p$. Moore cofiber sequences give the result for p^a, and Gabber rigidity [5, Theorem 1.4] makes the tower constant modulo ℓ^a for $\ell \neq p$. Suslin's calculation then gives (43).

Completion. The transition $\mathbb{Z}/p^{a+1}(k) \twoheadrightarrow \mathbb{Z}/p^a(k)$ is surjective, so the condensed Milnor sequence applied to (43) gives (44). Inverting p gives (45); compactness of a profinite test object supplies the uniform denominator.

Rationalization. For an extremally disconnected profinite S, applying [1, Theorem 4.14] to $\operatorname{Spf}(\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p}))$ gives

$$(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})\otimes\mathbb{Q})(S) \longrightarrow K(\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p})/p)\otimes\mathbb{Q} \longrightarrow \bigoplus_{i\geq 1}\Sigma^{2i} H\Big(B_{\mathrm{dR}}^{+}(\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p}))/\operatorname{Fil}^{i}\Big). \tag{48}$$

The middle term is $H\,\mathrm{LC}(S,\mathbb{Q})$. Applying Lemma 6.2 to the last term gives (46).

Odd finite-coefficient vanishing makes multiplication by every integer injective on each positive even homotopy object, while (46) makes its rationalization zero. Exactness of filtered colimits now gives (47). ☐

6.4. **The coherent arithmetic map.** Let

$$\underline{\partial_k} : \Sigma^{2k-1} H\underline{B_{\mathrm{dR}}^{+}/\operatorname{Fil}^{k} B_{\mathrm{dR}}^{+}}^{\mathrm{top}} \longrightarrow \Sigma^{2k} H\underline{\mathbb{Q}_p(k)}^{\mathrm{top}} \tag{49}$$

be the connecting morphism of (40) in the derived category of condensed Galois modules.

Proposition 6.5 (The coherent arithmetic gluing morphism)**.** *For every $k \geq 2$, the assembled weight-k component of the arithmetic comparison map is*

$$\lambda_k \underline{\partial_k}, \qquad \lambda_k \in \mathbb{Q}_p^{\times}, \tag{50}$$

in

$$D\Big(\operatorname{Mod}_{\mathbb{Z}[\underline{G_{\mathbb{Q}_p}}]}(\operatorname{Cond}_{\kappa}(\mathbf{Ab}))\Big).$$

The scalar compares the Kummer–Bott coordinate on (45) *with the period coordinate.*

Proof. For an extremally disconnected profinite S, CMM p-adic continuity and AMMN's continuous DGM square [1, (23)] give a natural cartesian diagram

$$\begin{tikzcd} \underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})(S) \arrow[r] \arrow[d, "\mathrm{tr}^{\mathrm{cts}}"'] & K(\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p})/p) \arrow[d, "\mathrm{tr}"] \\ \mathrm{TC}(\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p});\mathbb{Z}_p) \arrow[r] & \mathrm{TC}(\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p})/p;\mathbb{Z}_p). \end{tikzcd} \tag{51}$$

The ring $\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p})$ is p-henselian, and its quotient modulo p is semiperfect of Krull dimension zero. Thus [5, Corollary 6.9] applies even when $\operatorname{Cont}(S,\mathcal{O}_{\mathbb{C}_p})$ is not local.

The functorial splitting of [1, Corollary 6.23 and its proof] decomposes the actual rational TC morphism into Adams weights. Its weight-k part is the square of [1, Theorem 6.17], and [1, Theorem 7.7] expresses that square in period terms for $\mathrm{Cont}(S, \mathcal{O}_{\mathbb{C}_p})$. Under the identifications of Lemma 6.2, naturality for evaluation at $s \in S$ gives

$$\mathrm{ev}_s \circ \chi_{k,\mathrm{Cont}(S,\mathcal{O}_{\mathbb{C}_p})} = \chi_{k,\mathcal{O}_{\mathbb{C}_p}} \circ \mathrm{ev}_s .$$

These evaluations jointly detect maps between the corresponding function groups, so

$$\chi_{k,\mathrm{Cont}(S,\mathcal{O}_{\mathbb{C}_p})} = \mathrm{Cont}(S, \chi_{k,\mathcal{O}_{\mathbb{C}_p}}).$$

Remark 7.9 of [1] identifies $\chi_{k,\mathcal{O}_{\mathbb{C}_p}}$ with Fontaine's period map. Hence the degree-zero maps assemble to (40) on all extremally disconnected profinite test objects.

Sequence (40) is first assembled in condensed Galois modules, and (49) is its connecting morphism there. Evaluation at a fixed profinite S is not used to define this class; after evaluation, the degree-zero sequence may admit a noncanonical splitting in **Ab**.

The fiber inclusions in (51) identify the assembled weight-k arithmetic map with a scalar multiple of $\underline{\partial_k}$. The comparison in [1, Remark 7.9] identifies the period map at $\mathbb{C}_p$ with $d\log$ in weight one and, by multiplicativity, in higher weights. The scalar is nonzero because the cyclotomic trace is an equivalence in the positive degrees at issue. Naturality makes it independent of S. Every map is natural for $\mathrm{Cont}(S, \mathcal{O}_{\mathbb{C}_p}) \to \mathrm{Cont}(S \times G_{\mathbb{Q}_p}, \mathcal{O}_{\mathbb{C}_p})$ induced by (34); hence (50) holds with the asserted internal Galois action. $\square$

6.5. **Arithmetic fracture and assembly.**

Proof of Theorem 1.3. Fix $k \geq 2$. The p-local arithmetic-fracture square gives a fiber sequence in condensed spectra with internal Galois action:

$$\begin{aligned} \underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{(p)} &\longrightarrow \underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})^\wedge_p \oplus (\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \otimes \mathbb{Q}) \\ &\xrightarrow{\iota - c} \underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})^\wedge_p[1/p]. \end{aligned} \tag{52}$$

Proposition 6.4 and Proposition 6.5, applied in the Postnikov window $[2k-1, 2k]$, give an extension

$$0 \longrightarrow \Delta\big((\mathbb{Q}_p/\mathbb{Z}_p)(k)\big) \longrightarrow \pi^{\mathrm{cond}}_{2k-1}\big(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{(p)}\big) \longrightarrow \underline{B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}}^{\mathrm{top}} \longrightarrow 0. \tag{53}$$

Its Yoneda class is the class of (40), multiplied by λ_k and pushed out along

$$\underline{\mathbb{Q}_p(k)}^{\mathrm{top}} \longrightarrow \Delta\big((\mathbb{Q}_p/\mathbb{Z}_p)(k)\big).$$

Precomposing the rational $B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}$-coordinate in (52) by λ_k^{-1} changes the gluing map to $\underline{\partial_k}$. This scalar acts by a continuous internal Galois-equivariant automorphism of $\underline{B^+_{\mathrm{dR}}/\operatorname{Fil}^k B^+_{\mathrm{dR}}}^{\mathrm{top}}$. The pushout of the Fontaine sequence by the lattice quotient therefore gives

$$\pi^{\mathrm{cond}}_{2k-1}\big(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{(p)}\big) \cong \operatorname{coker}\left(\underline{\mathbb{Z}_p(k)}^{\mathrm{top}} \longrightarrow \underline{(B^+_{\mathrm{cris}})^{\varphi = p^k}}^{\mathrm{top}}\right). \tag{54}$$

For $\ell \neq p$, let

$$\rho : \underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow \Delta K(\mathcal{O}_{\mathbb{C}_p}/p)$$

be reduction to the special fiber. Gabber rigidity makes ρ/ℓ^a an equivalence for every $a \geq 1$. The positive K-groups of $\mathcal{O}_{\mathbb{C}_p}/p$ are torsion. Since the odd homotopy groups of $\Delta K(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a$ vanish, the Bockstein sequence in degree $2k+1$ gives

$$K_{2k}(\mathcal{O}_{\mathbb{C}_p}/p)[\ell^a] = 0.$$

Multiplication by ℓ^a is consequently an automorphism of the torsion group $K_{2k}(\mathcal{O}_{\mathbb{C}_p}/p)$, and hence $K_{2k}(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a = 0$. On the formal side, (47) gives $\pi_{2k}^{\mathrm{cond}}\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) = 0$. Thus the two Bockstein boundaries in the natural square

$$\begin{tikzcd}
\pi_{2k}^{\mathrm{cond}}\big(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})/\ell^a\big) \arrow[r, "\partial"] \arrow[d, "\sim"'] & \pi_{2k-1}^{\mathrm{cond}}\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})[\ell^a] \arrow[d, "\rho_*"] \\
\pi_{2k}^{\mathrm{cond}}\big(\Delta K(\mathcal{O}_{\mathbb{C}_p}/p)/\ell^a\big) \arrow[r, "\partial"] & \pi_{2k-1}^{\mathrm{cond}}\Delta K(\mathcal{O}_{\mathbb{C}_p}/p)[\ell^a]
\end{tikzcd}$$

are isomorphisms, and so is the right vertical map. Passing to the union over a, using Lemma 6.1, and applying the canonical primary projection on the torsion special-fiber group gives an internal Galois-equivariant retraction. Suslin's calculation identifies its image with

$$\bigoplus_{\ell\neq p}\Delta\big((\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\big).$$

For every positive integer s prime to p, odd finite-coefficient vanishing in (43) makes multiplication by s an epimorphism on $\pi_{2k-1}^{\mathrm{cond}}\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})$. The heart of condensed spectra is Grothendieck abelian, so filtered colimits are exact. Since $\mathbb{Z}_{(p)} = \mathrm{colim}_{(s,p)=1}\,\mathbb{Z}$, localization is an epimorphism

$$\pi_{2k-1}^{\mathrm{cond}}\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p}) \longrightarrow \pi_{2k-1}^{\mathrm{cond}}\big(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{(p)}\big)$$

whose kernel is the union of the subobjects killed by such s, namely the displayed prime-to-p torsion object. Together with localization, the special-fiber retraction is an isomorphism: a lift of a localized class is corrected by subtracting its retracted torsion component. Lemma 6.1 transports this decomposition to $\underline{K}^{\mathrm{nuc}}(\mathbb{C}_p)$. Hence there is a split exact sequence

$$\begin{aligned}0 \longrightarrow \bigoplus_{\ell\neq p}\Delta\big((\mathbb{Q}_\ell/\mathbb{Z}_\ell)(k)\big) &\longrightarrow \pi_{2k-1}^{\mathrm{cond}}\underline{K}^{\mathrm{nuc}}(\mathbb{C}_p)\\ &\longrightarrow \pi_{2k-1}^{\mathrm{cond}}\big(\underline{K}_{\mathrm{form}}(\mathcal{O}_{\mathbb{C}_p})_{(p)}\big) \longrightarrow 0.\end{aligned} \tag{55}$$

Substitution of (54) into (55) proves (3); Lemma 6.3 identifies the topology of its p-local summand. □

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